

Table S3. Image quality metrics in the Radar plot. Explanation and of the different image quality metrics and the calculated normalized polygon area (*A*) including the range of each metric in the Radar plot.

Metric	Brief description	Detailed description	What is “good”?	Axis range in this work
SSIM	Fidelity to reference	Structural Similarity Index measures similarity to a reference image by comparing local structure, contrast, and luminance. Here we use the sub-sampled image as a reference, implementing a masked version of SSIM.	higher values indicate better fidelity.	0-0.5
NIQE / PIQE / BRISQUE	Perceptual image quality	Natural Image Quality Evaluator is a no-reference perceptual image quality metric that measures deviation from natural image statistics.	lower values indicate better perceived quality. for	50-0
Resolution from FFT (r_{fft})	Finest detail size	Estimates the smallest resolvable feature size from the Fourier amplitude spectrum.	lower values correspond to higher spatial resolution.	10-0.74 nm
Salt & Pepper fraction	Impulse artifacts	Estimates the fraction of pixels exhibiting impulse-like intensity spikes relative to their local neighborhood.	higher values indicate more salt-and-pepper artifacts.	0-1
Roughness (P90)	Graininess	Quantifies high-frequency texture by measuring deviations from a Gaussian-smoothed version of the image. We used the 90th percentile of the roughness map so that the metric reflects the upper range of texture present in the image without being dominated by a small number of extreme pixels caused by noise or artifacts.	higher values indicate greater graininess or fine-scale fluctuations.	0.02-0
DOM	Edge sharpness	Difference of Medians (DOM) sharpness quantifies image sharpness based on local intensity transitions.	higher values indicate sharper features and edges.	0-1.1
Normalized polygon area (<i>A</i>)	Combined quality score	The ratio between the image specific polygon area and the area of an ideal polygon corresponding to the optimal value of every metric.	larger values indicate better overall performance across all included metrics.	0-1